

Lab 7

Dataset: Iris Dataset (`sklearn.datasets.load_iris()`)

Task 1: Dataset Exploration

Load the Iris dataset and answer the following:

1. How many samples and features are present in the dataset?
 2. List the feature names and target classes.
 3. Display the first five records.
 4. Display the class distribution.
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Task 2: Data Preparation

1. Split the dataset into training (80%) and testing (20%) sets using `random_state=42`.
 2. Briefly explain why the dataset is divided into training and testing sets.
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Task 3: Building a Decision Tree Classifier

1. Train a Decision Tree classifier using the default parameters.
 2. Predict the class labels for the test dataset.
 3. Evaluate the model using:
 - Accuracy Score
 - Confusion Matrix
 - Classification Report
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Task 4: Visualizing the Decision Tree

1. Visualize the trained Decision Tree using `plot_tree()`.
 2. From the visualization, identify:
 - Root node
 - Internal nodes
 - Leaf nodes
 - Maximum depth of the tree
 3. Which feature is selected for the first split? Explain why you think this feature was chosen.
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Task 5: Comparing Gini Index and Entropy

Train two Decision Tree models using the following criteria:

- `criterion='gini'`
- `criterion='entropy'`

Compare the models based on:

1. Accuracy
2. Tree depth
3. Number of leaf nodes
4. Root feature selected
5. Which criterion produces a simpler tree?

Summarize your observations in a table.

Task 6: Effect of Maximum Tree Depth

Train Decision Tree models using the following values of `max_depth`:

- 1
- 2
- 3
- 4
- None

Complete the following table.

Max Depth	Training Accuracy	Testing Accuracy	Tree Depth	Observation
1				
2				
3				
4				
None				

Answer the following:

1. Which model is underfitting?

2. Which model gives the best generalization?
 3. Which model is likely to overfit? Justify your answer.
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Task 7: Effect of `min_samples_split`

Train Decision Tree models using:

- 2
- 5
- 10
- 20

Complete the table below.

<code>min_samples_split</code>	Accuracy	Tree Depth	Number of Leaf Nodes	Observation
2				
5				
10				
20				

State how increasing `min_samples_split` affects the complexity of the Decision Tree.

Task 8: Effect of `min_samples_leaf`

Train Decision Tree models using:

- 1
- 2
- 5
- 10

Complete the following table.

<code>min_samples_leaf</code>	Accuracy	Tree Depth	Number of Leaf Nodes	Observation
1				
2				

5

10

Explain how changing `min_samples_leaf` influences overfitting.

Task 9

Answer the following questions.

1. What is the role of the **criterion** parameter in a Decision Tree?
2. How does `max_depth` influence underfitting and overfitting?
3. Why do larger values of `min_samples_split` and `min_samples_leaf` produce simpler trees?
4. Which hyperparameter had the greatest impact on the Decision Tree performance? Support your answer with observations from the experiments.
5. Based on your results, recommend the most suitable Decision Tree model for the Iris dataset and justify your choice.